

Cryptography - 31_10

🔗 Exercise 1

Prove or refute: An SKE scheme is Perfectly secret if and only if:

$\forall M \text{ over } \mathcal{M}, \forall c_0, c_1 \in \mathcal{C}$

$Pr[C = c_0] = Pr[C = c_1]$ with $C = Enc(k, M)$, and k is uniform

On the one hand, it seems that $(i) \implies (ii)$, where (i) is the statement in the exercise, and (ii) is $Pr[M = m] = Pr[M = m | C = c]$. In other words, if a scheme is perfectly secret, is it necessary that all cipher texts are equally likely? It's not true: Here is a counter example. Consider $\bar{\Pi} = (\bar{Enc}, \bar{Dec})$ such that $\bar{Enc}(k, m) = 0 || m \oplus k$, and $\bar{Dec}(k, b || c) = c \oplus k$, with $b \in \{0, 1\}$. So this is perfectly secret, satisfying (ii) , but does it also satisfy (i) ?

Now, take $c_1 \in \mathcal{C}$ to be $1 || \bar{c}$ and c_0 to be $0 || c$. Then the probability of the first is zero, and the one of the second is not zero. So we've reached the conclusion that

$$Pr[C = c_0] \neq Pr[C = c_1]$$

🔗 Exercise 2 (at home)

Prove that an SKE Π is perfectly secret if and only if $\forall$ adversaries (unbounded):

$$GAME_{\Pi, \mathcal{A}}^{one-time}(\lambda, 0) \equiv GAME_{\Pi, \mathcal{A}}^{one-time}(\lambda, 1)$$

🔗 Exercise 3

We are given $f_1 : \{0, 1\}^n \rightarrow \{0, 1\}^n$ and $f_2 : \{0, 1\}^n \rightarrow \{0, 1\}^n$ and we know that at least one is a OWF.

Design f a OWF using f_1, f_2 .

A proposal is given by $f(x) = f_1(f_2(x))$. Try to make the reduction:

We start by assuming f_2 is secure (?), and we build A_f .

Problem: $f_1(U_n)$ might have different distribution than $f_1(f_2(U_n))$.

Counter example:

Let $f_2(x) = 0^n \forall x \in \{0, 1\}^n$. Let $f_1(x) =$

$$= \begin{cases} f'(x) & \text{if } f \neq 0^n \\ 0^n & \text{if } x = 0^n \end{cases}$$

where f' is any OWF.

Another attempt:

$$f(x_1 || x_2) = f_1(x_1) || f_2(x_2)$$

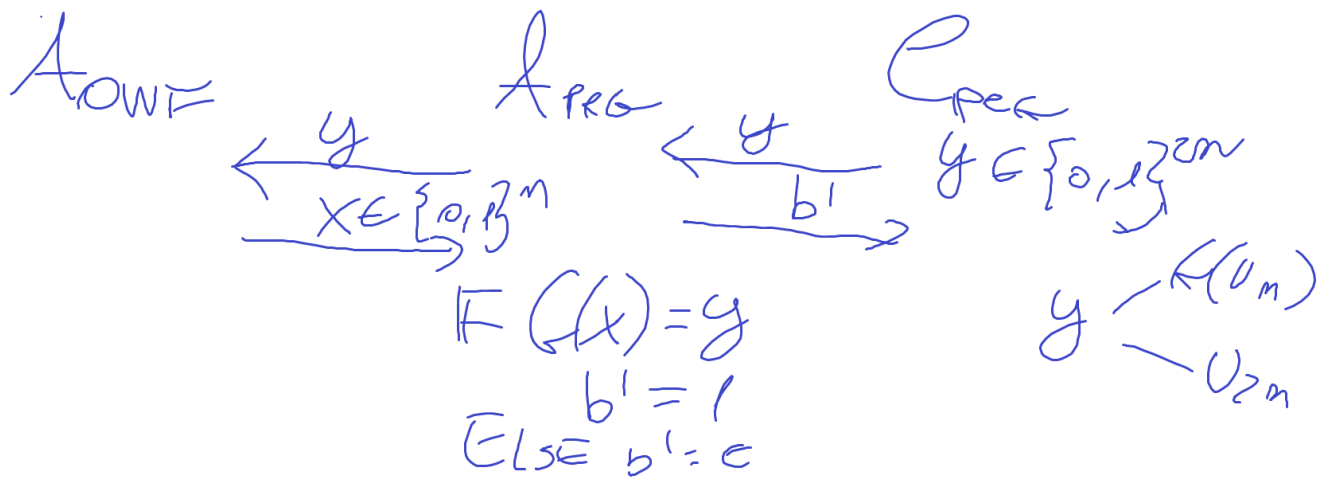
with $f : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n}$.

This works.

Work it out at home.

🔗 Exercise 4

Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a PRG. Show G is a OWF.



Now, we need to compute

$$Pr[b' = 1 : y = G(U_n)]$$

so the probability that $b' = 1$ when y comes from a PRG, and

$$Pr[b' = 1 : y \leftarrow U_{2n}]$$

The first one is $\geq \mathcal{E}(n)$, where $\mathcal{E}(n) = Pr[\mathcal{A}(G(U_n)) \text{ wins}] \geq \frac{1}{poly}$.

The second one is $\leq \frac{2^n}{2^{2n}} = 2^{-n}$.

🔗 Exercise 5

Let $F'_k(x) = F_k(0||x)||F_k(x||1)$. Assume $F_k : \{0, 1\}^n \rightarrow \{0, 1\}^n$ a PRF. And $F'_k : \{0, 1\}^{n-1} \rightarrow \{0, 1\}^{2n}$.
Is F' a PRF?

No. Look:

$$F'_k(0^{n-1}) = F_k(0^n)||F_k(0^{n-1}||1). \text{ Also}$$

$$F'_k(0^{n-2}||1) = F_k(0^{n-1}||1)||F_k(--)$$

Then we have two queries where the first half of a query is equal to the second half of another query.

This gives immediately a distinguisher.

🔗 Exercise 6

Show CBC-MAC is not UFCMA for VIL.

We need to build an adversary for UCFMA game and a challenger for UFCMA.

The adversary can make queries, and works just like in UFCMA game.

Now, what is $\text{CBC-MAC}_k(m_1)$? It's $F_k(m_1)$. For $m_1||m_2$? It's $F_k(F_k(m_1) \oplus m_2)$.

Now the forge from A is going to be over $m^* = m_1||m_2$.

So, the queries are going to be m_1 that is going to give us τ_1 . The second query is $\tau_1 \oplus m_2$, that will give us τ^* , so we've successfully forged for VIL.

🔗 Exercise 7

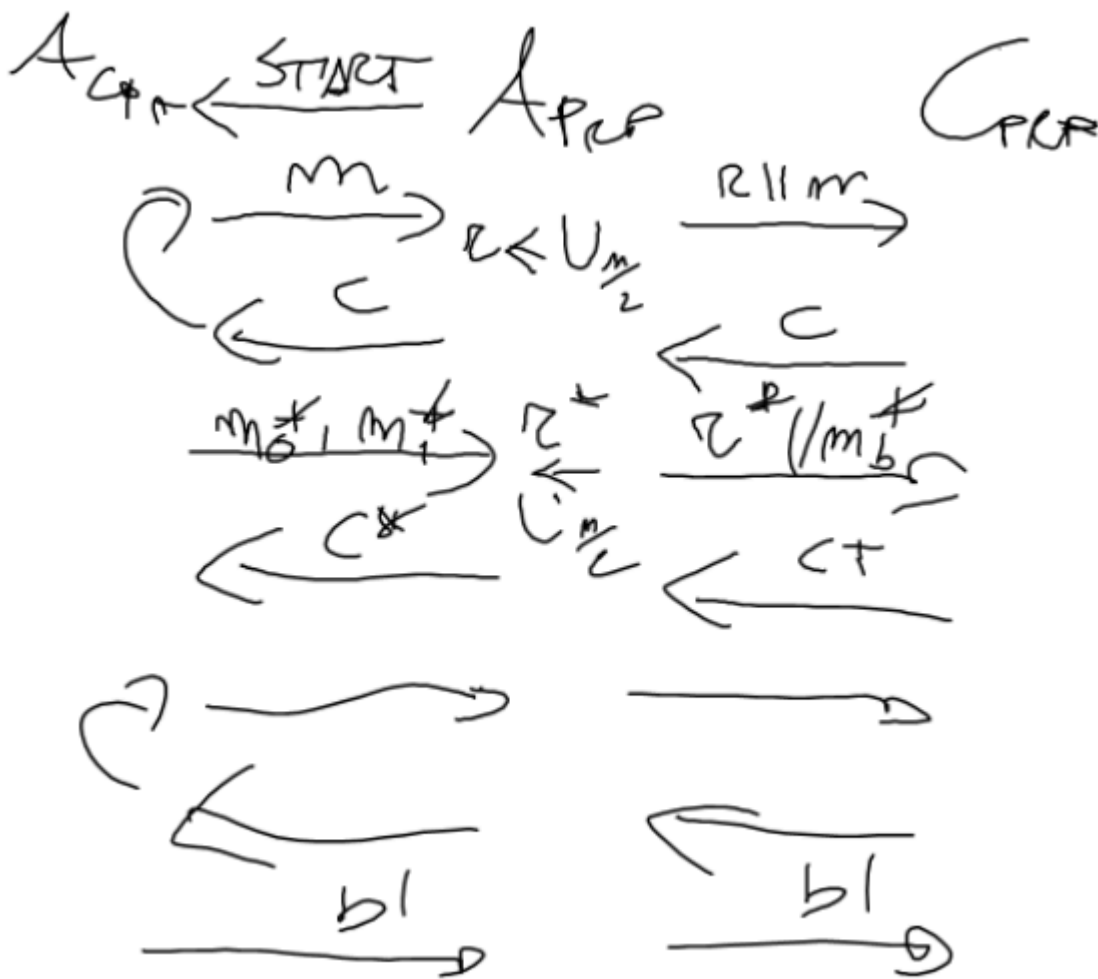
Let $\mathcal{F} = \{F_k\}$ a PRP.

Define (Enc, Dec) such that

$$Enc(k, m) = F_k(r||m), \text{ with } r \leftarrow U_{\frac{n}{2}}, \text{ and } m \in \{0, 1\}^{n/2}.$$

$$Dec(k, c) \text{ outputs } m \text{ such that } r||m = F_k^{-1}(c)$$

Prove it is CPA secure.



Then $G(\lambda, b)$: the CPA game with bit $b \in \{0, 1\}$

$H(\lambda, b)$: The CPA game with $F_k \rightarrow R$ a random permutation

By the above reduction:

$$\implies G(\lambda, b) \approx_c H(\lambda, b) \quad \forall b \in \{0, 1\}$$

So it remains to prove that $H(\lambda, 0) \approx_c H(\lambda, 1)$, so that we can prove $G(\lambda, 1) \approx_c G(\lambda, 0)$.

In $H(\lambda, b)$: $c^* = R(r^* || m_b^*)$.

The CPA queries: $c_i = R(r_i || m_i)$

We are going to have event BAD: the event that r^* equals one of the r_i 's.

Conditioning on $\overline{BAD}$, c^* is uniform.

$$\implies H(\lambda, 0) \equiv H(\lambda, 1).$$

$$\text{Then } \Pr[BAD] \leq q(\lambda) \cdot 2^{-\frac{n}{2}}$$

with $q(\lambda) = \text{poly}(\lambda) = \# \text{CPA queries}$